

# Equation Sheet PHYS 241

Elementary charge:  $1.602 \times 10^{-19} \text{ C}$

Mass of a proton:  $1.67 \times 10^{-27} \text{ kg}$

Mass of an electron:  $9.11 \times 10^{-31} \text{ kg}$

$$\vec{F} = q\vec{E} \quad Q(t) = C\epsilon(1 - e^{-t/\tau_{RC}}) \quad \epsilon_o = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \quad k = 8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} = \frac{1}{4\pi\epsilon_o} \quad \mu_o = 4\pi \times 10^{-7} \frac{\text{T} \cdot \text{m}}{\text{A}} \quad g = 9.81 \frac{\text{m}}{\text{s}^2} \quad c = 3 \times 10^8 \text{ m/s}$$

$$\vec{F} = \frac{q_1 q_2}{4\pi\epsilon_o r^2} \hat{r} \quad Q(t) = C\epsilon e^{-t/\tau_{RC}} \quad I = \frac{\epsilon}{R} e^{-t/\tau_{RC}} \quad \text{Resistors in Parallel: } \frac{1}{R_{equiv}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \quad \text{Capacitors in Series: } \frac{1}{C_{equiv}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

$$\vec{E} = \frac{q}{4\pi\epsilon_o r^2} \hat{r} \quad I = \frac{\epsilon_o}{R} e^{-t/\tau_{RL}} \quad \vec{F} = q\vec{v} \times \vec{B} \quad \text{Resistors in Series: } R_{equiv} = R_1 + R_2 + \dots \quad \text{Capacitors in Parallel: } C_{equiv} = C_1 + C_2 + \dots$$

$$V = \frac{kq}{r} \quad I = \frac{\epsilon_o}{R} (1 - e^{-t/\tau_{RL}}) \quad \epsilon = -\frac{d\Phi}{dt} \quad U = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$V = \frac{Q}{C} \quad \tau_{RC} = RC \quad \epsilon = \oint_{Loop} \vec{E} \cdot d\vec{l} \quad \omega = 2\pi f \quad \text{phase difference: } \frac{\delta}{2\pi} = \frac{\Delta r}{\lambda}$$

$$C = \frac{\epsilon_o A}{d} \quad u = \frac{1}{2} \epsilon_o E^2 \quad \vec{\mu} = NIA\hat{n} \quad X_C = \frac{1}{\omega C} \quad \text{two slits (constructive interference): } d \sin \theta = m\lambda$$

$$E_x = -\frac{dV}{dx} \quad \epsilon = \kappa \epsilon_o \quad \vec{\tau} = \vec{\mu} \times \vec{B} \quad X_L = \omega L \quad \text{two slits (destructive interference): } d \sin \theta = (m - \frac{1}{2})\lambda$$

$$E_y = -\frac{dV}{dy} \quad R = \rho \frac{L}{A} \quad L = \mu_o n^2 A l \quad \omega_{resonance} = \frac{1}{\sqrt{LC}} \quad \text{single slit (destructive interference): } d \sin \theta = m\lambda$$

$$E_z = -\frac{dV}{dz} \quad P = IV \quad U = -\vec{\mu} \cdot \vec{B} \quad I_{rms} = \frac{1}{\sqrt{2}} I_{peak} \quad \oint_{Curve} \vec{B} \cdot d\vec{l} = \mu_o (I + I_d) \quad I_d = \epsilon_o \frac{d\phi_E}{dt} \quad c = \frac{1}{\sqrt{\epsilon_o \mu_o}} \quad E = cB$$

$$\alpha = \frac{(\rho - \rho_o) / \rho_o}{T - T_o} \quad \oint_{Surface} \vec{B} \cdot d\vec{A} = 0 \quad I_{rms} = \frac{V_{rms}}{R} \quad \text{momentum} = U/c \quad \text{radiation pressure} = I/c \quad \text{intensity} = u_{ave} c$$

$$\Phi = \int_S \vec{E} \cdot \hat{n} dA = \frac{Q}{\epsilon_o} \quad \oint_{Loop} \vec{B} \cdot d\vec{l} = \mu_o I_{enclosed} \quad I_{rms} = \frac{V_{rms}}{X_C} \quad \text{intensity} = \frac{E_o B_o}{2\mu_o} \quad \text{intensity} = \text{power/area}$$

$$\Delta V = V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{l} \quad \vec{B} = \frac{\mu_o}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} \quad P_{ave} = I_{rms}^2 R \quad E = hf \quad h = 6.6 \times 10^{-34} \text{ Js} \quad v = c/n \quad \theta_i = \theta_i' \quad I = \left( \frac{n_1 - n_2}{n_1 + n_2} \right)^2 I_o$$

$$dU = -\vec{F} \cdot d\vec{l} \quad U = \frac{1}{2} LI^2 \quad I_{peak} = \frac{V_{peak}}{R} \quad n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad I = I_o \cos^2 \theta \quad \frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad \text{mirror: } f = r/2$$

$$dV = \frac{dU}{q_o} = -\vec{E} \cdot d\vec{l} \quad \Phi = \int_S \vec{B} \cdot d\vec{A} \quad I_{peak} = \frac{V_{peak}}{X_C} \quad m = \frac{y'}{y} = -\frac{s'}{s} \quad \frac{n_1}{s} + \frac{n_2}{s'} = \frac{n_2 - n_1}{r} \quad \frac{1}{f} = (n - 1) \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$L = \frac{\Phi}{I} \quad I_{peak} = \frac{V_{peak}}{X_L}$$

## Equation Sheet PHYS 241

| Mirror Type | Object Location | Image Location                      | Image Type | Image Orientation          | Sign of $f, r, s', m$  |
|-------------|-----------------|-------------------------------------|------------|----------------------------|------------------------|
| Plane       | Anywhere        | Opposite side of mirror from object | Virtual    | Same orientation as object | $\infty, \infty, -, +$ |
| Concave     | Inside F        | Opposite side of mirror from object | Virtual    | Same orientation as object | $+, +, -, +$           |
| Concave     | Outside F       | Same side as object                 | Real       | Inverted                   | $+, +, +, -$           |
| Convex      | Anywhere        | Opposite side of mirror from object | Virtual    | Same orientation as object | $-, -, -, +$           |

| Lens Type  | Object Location | Image Location                   | Image Type | Image Orientation | Sign of $f, s', m$ |
|------------|-----------------|----------------------------------|------------|-------------------|--------------------|
| Converging | Inside F        | Same side as object              | Virtual    | Same as object    | $+, -, +$          |
| Converging | Outside F       | Side of lens opposite the object | Real       | Inverted          | $+, +, -$          |
| Diverging  | Anywhere        | Same side as object              | Virtual    | Same as object    | $-, -, +$          |